Project Gizmo: Subspace-Constrained Quantum Portfolio Selection

Eliminating QAOA Penalty Dilution via Excitation-Preserving Mixers & $\mathcal{O}(N)$ SWAP Networks

Team Gizmo
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The Core Problem: Cardinality-Constrained Portfolio Selection

Markowitz Mean-Variance Formulation (1952):

$$\min_{\mathbf{x}\in\{0,1\}^N} \mathcal{C}(\mathbf{x}) = \frac{q}{2}\mathbf{x}^T\boldsymbol{\Sigma}\mathbf{x} - \boldsymbol{\mu}^T\mathbf{x}$$

subject to the exact budget/cardinality constraint:

$$\sum_{i=0}^{N-1} x_i = K \quad \text{(Pick exactly } K \text{ out of } N \text{ assets)}$$

- $\boldsymbol{\mu} \in \mathbb{R}^N$: Asset expected annual returns.
- $\boldsymbol{\Sigma} \in \mathbb{R}^{N\times N}$: Return covariance matrix.
- $q > 0$: Risk-aversion coefficient.
- **Complexity:** NP-hard binary quadratic optimization.

The Combinatorial Reality

- Total search space: 2^N states.
- Feasible asset combinations: $\binom{N}{K}$.
- **Example ($N = 16, K = 4$):**
  - Total states: $65,536$
  - Feasible states: $1,820$
  - Valid fraction: only 2.78%
- How does standard gate-model quantum optimization handle this constraint?

The Conventional Approach & Its Catastrophic Failure

The Standard Textbook Recipe: Map the constraint to a soft penalty term:

$$H_{\text{total}} = H_C + \lambda \left(\sum_{i=0}^{N-1} x_i - K \right)^2, \quad H_M = \sum_{i=0}^{N-1} X_i, \quad |\psi_0\rangle = |+\rangle^{\otimes N}$$

Failure 1: The Penalty Parameter Dilemma

- λ **too small:** Quantum state collapses into unconstrained space; $> 60\%$ of measured shots violate the budget ($\sum x_i \neq K$).
- λ **too large:** Penalty canyons dominate the cost landscape; optimizer ignores variance $\boldsymbol{\Sigma}$ and returns $\boldsymbol{\mu}$.

Failure 2: Exponential State Dilution

- Transverse mixer $\sum X_i$ breaks Hamming weight conservation.
- Amplitudes leak indiscriminately into the 2^N unconstrained space.
- On noisy NISQ hardware, $> 95\%$ of experimental shot budget is wasted on illegal portfolios.

The Gizmo Architecture: Complete Subspace Confinement

Core Thesis: Stop penalizing invalid configurations. Confine the Hilbert space.

1. Deterministic Dicke Initialization

- Initialize in the symmetric superposition:

$$|D_K^N\rangle = \binom{N}{K}^{-1/2} \sum_{\mathrm{wt}(\mathbf{x})=K} |\mathbf{x}\rangle$$

- Synthesized via an inductive $\mathcal{O}(NK)$ controlled-Givens rotation ladder (Bärtschi & Eidenbenz).
- Zero classical statevector allocation.

2. Excitation-Preserving XY-Mixer

- Replace X_i with hopping interactions:

$$H_{XY} = \frac{1}{2} \sum_{i=0}^{N-2} (X_i X_{i+1} + Y_i Y_{i+1})$$

- Commutes with total excitation:

$$\left[H_{XY}, \sum_{i=0}^{N-1} Z_i \right] = 0$$

- **Guarantees 100% feasible measurement samples.**

Mathematical Rigor: Commutation Proof & Subspace Invariance

Analytical Commutator Derivation on Edge (i, j):

$$\begin{aligned}[X_iX_j + Y_iY_j, Z_i + Z_j] &= [X_iX_j, Z_i] + [X_iX_j, Z_j] + [Y_iY_j, Z_i] + [Y_iY_j, Z_j] \\ &= (-2iY_iX_j) + (-2iX_iY_j) + (2iX_iY_j) + (2iY_iX_j) \\ &= 0\end{aligned}$$

Direct Consequence

For any evolution angle $\beta \in \mathbb{R}$:

$$\left[e^{-i\beta H_{XY}}, \sum_{i=0}^{N-1} Z_i \right] = 0$$

The unitary $U_{XY}(\beta)$ leaves the eigenspaces of total Hamming weight strictly invariant.

Elimination of Hyperparameters

- Penalty multiplier λ is completely eliminated.
- Classical outer optimizer (COBYLA) searches strictly within the $\binom{N}{K}$ financial subspace.
- Every measurement shot is a valid allocation.

Heavy-Hex Compilation: Linear $\mathcal{O}(N)$ Alternating SWAP Network

- **The Bottleneck:** Return covariance $\mathbf{\Sigma}$ is dense $\implies \binom{N}{2}$ all-to-all $Z_i Z_j$ interactions.
- Mapping K_N onto IBM heavy-hex (degree ≤ 3) via standard heuristic transpilation creates dozens of uncontrolled SWAPs.

Linear Alternating SWAP Network

- Arrange N qubits on a 1D nearest-neighbor chain.
- In each stage: apply $R_{ZZ}(2\gamma J_{q_a,q_b})$ on adjacent physical wires, followed by SWAP.
- Alternating even/odd physical pairs executes all $\frac{N(N-1)}{2}$ interactions in **exactly N stages ($\mathcal{O}(N)$ depth)**.

Permutation Unrolling

- After N stages, physical wire p holds logical asset $\sigma(p)$.
- **The Innovation:** Measure physical wire p directly into classical register bit $\sigma(p)$.
- **Zero inverse-SWAP gate overhead!**

Empirical Benchmark: S&P 500 Portfolio ($N = 8, K = 3$, 4,096 Shots)

Metric	Naive QAOA ($\lambda = 2.5$)	Gizmo Subspace	Gizmo SWAP-Net	Ground Truth
Total Shots	4,096	4,096	4,096	56
Feasible Shots ($\sum x_i = 3$)	1,636	**4,096**	**4,096**	56
Feasibility Ratio	**39.94%**	**100.00%**	**100.00%**	100.00%
Min Energy Found	-0.956407	**-0.956407**	**-0.956407**	**-0.956407**
Circuit Depth	15	20	**22 (Bounded)**	—

Financial Validation

- Selected Assets: **AAPL, XOM, JNJ** (Bitstring 00110001).
- Expected Return: **34.99%** Volatility: **14.40%**
- **Optimal Sharpe Ratio: 2.2222** (vs. $r_f = 3.0\%$)

The Waste in Naive QAOA

- Naive QAOA wasted **60.06% of total measurement shots** on invalid states ($K \neq 3$).
- Gizmo achieved **zero wasted shots** and recovered the exact combinatorial minimum energy.

The Exponential Scaling Wall: Subspace Confinement vs. Unconstrained Diffusion

Assets (N)	Budget (K)	Naive Space $\frac{\binom{N}{K}}{2^N}$	Gizmo Subspace
6	2	23.44%	**100.00%**
8	2	10.94%	**100.00%**
10	2	4.39%	**100.00%**
12	3	5.37%	**100.00%**
14	3	2.22%	**100.00%**
16	4	2.78%	**100.00%**
20	5	**1.48%**	**100.00%**

Exponential Sampling Advantage

- At $N = 16$, naive QAOA wastes **97.2**% of shots.
- At $N = 20$, naive QAOA wastes **98.5**% of shots.
- **Gizmo Advantage Factor:**

$$\frac{2^{20}}{\binom{20}{5}} \approx \mathbf{67.6}\times$$

- Subspace confinement is mathematically non-negotiable for scale.

Epistemic Honesty: Why Not Just Use Classical MIP / Gurobi?

Intellectual Honesty Check

For standard quadratic binary programs with $N \leq 50$, classical mixed-integer solvers (Gurobi, CPLEX, SCIP) reach global optimality in milliseconds. We make no false claims of beating classical MIPs on toy quadratic models.

Where Quantum Advantage Actually Resides: Higher-Order Binary Optimization (HUBO)

- **Nonlinear Market Impact:** Piecewise cubic transaction costs ($c_i x_i^{3/2}$) where classical convex relaxations loosen.
- **Higher-Order Tail Risk:** Asymmetric downside risk incorporating co-skewness ($S_{ijk} x_i x_j x_k$) and co-kurtosis ($K_{ijkl} x_i x_j x_k x_l$) risk tensors.
- **The Frontier:** Classical branch-and-cut faces worst-case exponential tree exploration under HUBO, whereas gate-model quantum superposition naturally executes multi-qubit phase rotations.

Conclusion & Hackathon Deliverables

Technical Achievements

1. **100.00% Feasibility:** Eliminated soft-penalty tuning; verified zero constraint violations.
2. **Polynomial Dicke Ladder:** Replaced classical statevector allocations with $\mathcal{O}(NK)$ Givens rotations.
3. **Hardware-Aware SWAP Network:** $\mathcal{O}(N)$ depth all-to-all execution with zero-overhead classical permutation decoding.
4. **Rigorous Empirical Proof:** Verified exact combinatorial ground state on real S&P 500 historical data.

Project Links & Artifacts

- **Team Name:** Gizmo
- **Lead / Contact:** Akhilesh Varma
- **Email:** `akhverm@gmail.com`
- **Repository:** `https://github.com/ak495867/WQU-Gizmo`
- **Executed Notebook:** `gizmo_portfolio_showcase.ipynb`
- **Formal Paper:** `docs/gizmo_paper.pdf`